Chapter 9 Problem Set

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1. Given two positive integers a and b such that $b = a + 2$, the expression $\frac{1}{a} + \frac{1}{b}$ can be used to generate a Pythagorean triple. Specifically, the numerator is the first term of a Pythagorean triple, and the denominator is the second term.

Simplifying $\frac{1}{a} + \frac{1}{a+2}$, we get $\frac{2a+2}{a^2+2a}$. And substituting $m = a+1$ and $n = 1$ into the parameterization of Pythagorean triples, we get $(a^2+2a, 2a+2, a^2+2a+2)$. The fraction's numerator is the middle term of the Pythagorean triple, and the denominator is the first term.

2. Using the parameterization of Pythagorean triples, we see that $2(c-a) = 2(n^2 + m^2 - n^2 + m^2) = 2(2m^2) = 4m^2 = (2m)^2$.

3. The parameterization for $a^2 + 2b^2 = c^2$ is $(m^2 - 2n^2, 4mn, m^2 + 2n^2)$ with relatively prime integers m and n. First, let's prove that all solutions of that equation can be parameterized this way.

The second term in the parameterization is $4mn$, so b must be even. Therefore, we can write $b = 2d$. We know that $b^2 = c^2 - a^2$, so $4d^2 = (c - a)(c + a)$, or $d^2 = \left(\frac{c-a}{2}\right)\left(\frac{c+a}{2}\right)$. Since $\frac{c-a}{2}$ and $\frac{c+a}{2}$ are relatively prime, both of them are perfect squares. Since we have $d^2 = \frac{b^2}{4} = mn$, we can set $m = \sqrt{\frac{c-a}{2}}$ and $n = \sqrt{\frac{c+a}{2}}$, and both are integers because the radicands are perfect squares. Therefore, we have found values for m and n that satisfy the condition.

Now, we need to prove that all possible pairs (m, n) where m and n are relatively prime result in relatively prime values of a, b, and c that satisfy $a^2 + 2b^2 = c^2$. We can easily verify that $(m^2 - 2n^2)^2 + (4mn)^2 = (m^2 + 2n^2)^2$, so showing that $\gcd(m^2 - 2n^2, m^2 + 2n^2) = 1$ will suffice.

Let's say a prime p divides $m^2 - 2n^2$ and $m^2 + 2n^2$. That means $m^2 - 2n^2 = pk$ and $m^2 + 2n^2 = pl$. Subtracting both equations, we get $4n^2 = p(l - k)$. That means either p and $l - k$ are both perfect squares or $p = l - k$. The first is impossible because p is prime, and the second is also impossible because $p(l - k) = (2n)^2$, so $p = 2n$, which can't be true because p is prime. Therefore,

no such prime p exists, so the gcd of the first and third terms is 1.

4. We can factorize x^3+y^3 as $p^n = (x+y)(x^2-xy+y^2)$. That means both factors are powers of p. That means $x+y = p^k$, so $p^n = p^k((x+y)^2-3xy) = p^k(p^{2k}-3xy)$, so $p^{n-k} = p^{2k}-3xy$. That means $3xy \equiv 0 \pmod{p}$. So either $p = 3$, $p = x$, or $y = p$. These cases guarantee that there exist integers x, y, and n such that $p^n = x^3+y^3$.

5. When $c = 65$, there are the triples: $(16, 63, 65)$ and $(33, 56, 65)$.
When $c = 85$, there are the triples: $(13, 84, 85)$ and $(36, 77, 85)$.
When $c = 145$, there are the triples: $(17, 144, 145)$ and $(24, 143, 145)$.

6. There are at least 4 different primitive Pythagorean triples when $c = 1105$:

$$(47, 1104, 1105), (744, 817, 1105), (576, 943, 1105), (264, 1073, 1105).$$

8. Let's assume that (x, y, z) is the smallest possible solution to $x^4-y^4 = z^2$. We can factor that equation as $(x^2+y^2)(x^2-y^2) = z^2$. Since x^2+y^2 and x^2-y^2 are relatively prime, each of them is a perfect square, so we have $x^2+y^2 = a^2$ and $x^2-y^2 = b^2$. That means there exist integers m and n such that $x = m^2-n^2$, $y = 2mn$, and $a = m^2+n^2$. I didn't know how to continue from there.

9. Since f takes in natural numbers and returns natural numbers, to follow the criteria, it can never decrease. Also, f can only involve addition, multiplication, and exponentiation with n because otherwise, it could return values that aren't integers. If $f(n) = n+k$ where k is an integer greater than or equal to 1, that means $f(n+1) = n+k+1 \not> n+2k = f(n+k) = f(f(n))$. If $f(n) = kn$, then $f(n+1) = kn+n \not> k^2n = f(kn) = f(f(n))$. And finally, if $f(n) = n^k$, then $f(n+1) = (n+1)^k \not> n^{k^2} = f(n^k) = f(f(n))$. Since you can't add, multiply, or exponentiate anything to n and it can't decrease, $f(n) = n$.

10. It suffices to prove that $\sum_{i=1}^{n} a_i^3 \neq b^3$ for any positive integer n and different integers a_i, since if the volumes of the smaller cubes don't add up to the largest one, they won't fit inside it.

To prove this, we can rewrite $\sum_{i=1}^{n} a_i^3$ as $\left(\frac{n(n+1)}{2}\right)^2$. That expression needs to be a perfect cube. For that to be true, either one of n or $n+1$ must be a perfect cube and the other must be twice a perfect cube. The only possible remainders when a perfect cube is divided by 7 are 0, 1, and 6, and when divided by 9, they are 0, 1, and 8.

If k is a perfect cube, then $k \equiv 0, 1, 6 \pmod{7}$, so $k+1 \equiv 1, 2, 0 \pmod{7}$ respectively. The only case that works is when $k \equiv 6 \pmod{7}$. Considering $k \pmod{9}$ as well shows that 6 cannot be a remainder, because then $k+1 \equiv 7 \pmod{9}$, which can't be twice a perfect cube. Therefore, it is impossible for k

to be a perfect cube and for $k+1$ to be twice a perfect cube.

If $k+1$ is a perfect cube, then $k \equiv 0, 5, 6 \pmod 7$. Since k is twice a perfect cube, $k \equiv 0 \pmod 7$ is the only possibility. From there, I couldn't figure out what to do to complete the proof.